

SignSpeak: Smart Gloves for Deaf & Mute Communication

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Abstract. Deaf and mute people tend to get a barrier in communicating with people who are not familiar with sign language. To overcome this difficulty we propose a smart wearable glove system that is able to perform real-time Indian Sign Language (ISL) gesture-to-speech translation. The prototype is the combination of flex sensor for bending finger detection, MPU6050 accelerometer-gyroscope module for hand orientation & motion dynamics capturing. Sensor data is processed by ESP32 - S3 microcontroller using GRU (Gated Recurrent Unit) neural network trained on the sequential data of the gesture for the accurate recognition of the gesture. The trained model is optimized and ported into ESP32-S3 in TensorFlow Lite format to get the real-time inference with minimum latency. Through recognition, the corresponding audio that has been pre-recorded is retrieved from a microSD card and played through a DFPlayer Mini audio module with a mini speaker. Compared to current mobile applications and bulky devices, the proposed system is lightweight and portable as well as cost effective and can therefore be used in day-to-day life in schools and workplaces, healthcare and public spaces. This solution encourages accessibility, inclusivity, and independence for the speech and hearing impaired community and their ability to communicate.

Keywords: Sign language translation · wearable technology · Indian Sign Language (ISL) · ESP32-S3 · GRU · DFPlayer Mini · sensor gloves · embedded AI

1 Introduction

Communication is basic to human interaction, but to millions of people affected by hearing and speech disabilities it is a struggle on a daily basis. Indian Sign Language (ISL) is an important mode of communication, however, due to a lack of understanding among the general population this develops into barriers both socially and in the place of work. To solve this issue, SignSpeak, a smart glove with a set of sensors, is proposed in this project to recognize the ISL gestures and transfer them into speech in real time. Through the use of wearable technology, the system provides an easy and natural mode of communication, which can

help create a more inclusive environment and also allow persons to communicate confidently in a variety of scenarios such as in classrooms, workplaces, and public spaces.

The project is aimed at building a working prototype of smart gloves that can identify common ISL hand gestures and translate them into spoken words. Flex sensors placed on the fingers detect the bending movements which are computed by a microcontroller and converted to voice output through a speech synthesis module. The present design focuses on real-time performance, portability and affordability, as well as a basic ISL vocabulary. Future developments may add more comprehensive gesture recognition, add wireless connectivity for smartphones, and add multi-language support to make the solution more versatile and effective so that it can be adopted more widely in residences, schools, workplaces, and public spaces.

Table 1. Prevalence of Hearing Impairment in India by Age Group

Age Group	Condition	Prevalence (%)
0–14 years	Childhood hearing loss	6.6–16.5
15–59 years	Hearing impairment	6–26.9
60+ years	Age-related hearing loss	16.7–40
All ages	Disabling hearing loss	4.5–18.3

2 Related Work

Several researchers have explored the development of sensor-based gloves and machine learning approaches for sign language translation.

Leiva et al. [1] proposed a real-time gesture recognition system using the MPU6050 accelerometer–gyroscope sensor combined with traditional machine learning models such as SVM and KNN. Their work demonstrated the effectiveness of sensor-based motion capture for sign language recognition, but it was limited by small datasets and lacked audio output integration.

Hussain et al. [2] developed iSignNet, an IoT-based framework that utilized flex sensors and Arduino Uno to translate hand gestures into text on an LCD, while a DF Mini Player module generated speech. Although inexpensive and novel, the system lacked scalability and advanced deep learning support for dynamic gestures.

Tippannavar et al. [3] designed a glove with four flex sensors and an accelerometer, interfaced with an Arduino Mega 2560 to convert gestures into audio. While effective for static gestures, the system struggled with sequential or complex gesture recognition due to its reliance on basic rule-based mapping.

Azman et al. [4] introduced a hybrid glove–mobile application model that recognized gestures and displayed results via an Android app. This improved accessibility but depended on an external device, reducing portability.

More recent works, such as Kim et al. [8], applied Long Short-Term Memory (LSTM) networks to classify sequential hand motion data, achieving improved accuracy for dynamic gestures. Similarly, Liang et al. [9] explored deep learning approaches using convolutional networks (CNNs) for complex sentence-based sign recognition. These studies highlight the shift toward deep learning-based approaches for better accuracy.

While these solutions achieved promising results, most suffered from limited vocabulary, lack of portability or absence of real time speech output. Our proposed system addresses these gaps by combining flex sensors, MPU6050 Motion sensing using ESP32-S3 with GRU model processing used in TensorFlow Lite, as well as a DFPlayer Mini and real-time gesture-to-speech conversion speaker. This integration ensures low cost, portable and accurate real-time communication assistance to the deaf and mute community.

3 System Architecture

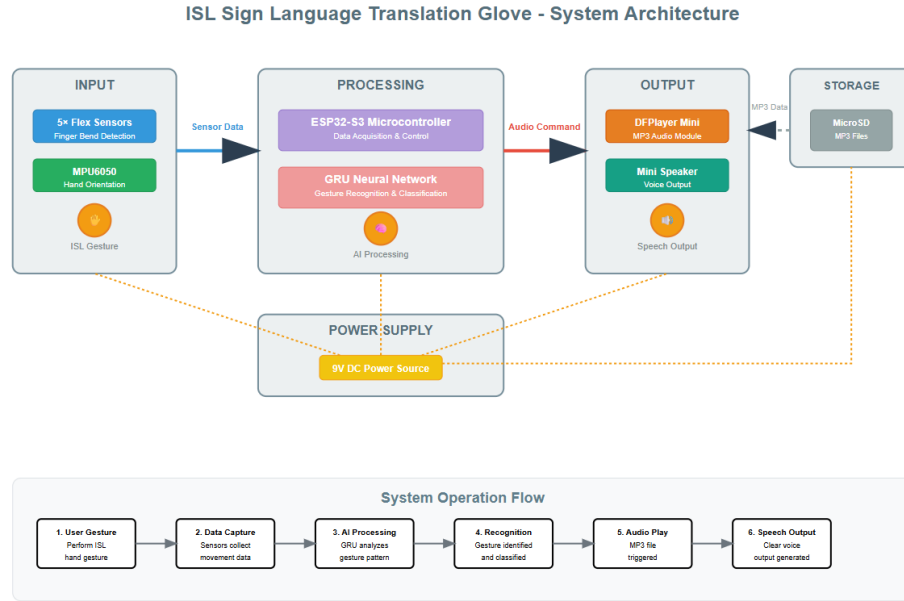


Fig. 1. System architecture of ISL to Speech Translation Gloves.

The proposed system architecture is designed as a wearable, real-time gesture recognition and speech synthesis system. It integrates multi-modal sensing, embedded deep learning inference, and audio playback into a compact device. The architecture consists of four layers: **Input Layer (sensors)**, **Processing**

Layer (ESP32-S3 with GRU inference), Output Layer (DFPlayer Mini and speaker), and Power Management Layer.

3.1 Input Layer – Sensor Subsystem

Five flex sensors are embedded along each finger to measure bending through resistance variation proportional to finger angle. Analog signals are routed to the ESP32-S3 ADC pins, with an external ADS1115 16-bit ADC used for higher precision multi-channel acquisition.

An MPU6050 accelerometer-gyroscope module captures wrist orientation, tilt, and dynamic motion. The sensor communicates via the I²C interface, providing tri-axial acceleration and angular velocity measurements.

Together, the sensor subsystem provides finger bend angles, hand orientation, and temporal motion dynamics forming a multidimensional time-series sequence for gesture modeling.

3.2 Processing Layer – Embedded GRU Inference

The ESP32-S3 microcontroller performs data acquisition, preprocessing, and real-time inference. Sensor streams from the flex sensors and MPU6050 are filtered, normalized, and formatted into sequential input windows for neural network processing.

A Gated Recurrent Unit (GRU) model is employed to capture temporal dependencies in gesture sequences. The model is trained offline using TensorFlow and converted to TensorFlow Lite format for deployment. The quantized model is embedded into the ESP32-S3 firmware and executed using TensorFlow Lite Micro, enabling continuous on-device inference with low latency.

Each predicted gesture class is mapped to a predefined audio file for speech output.

3.3 Output Layer – Audio Subsystem

The DFPlayer Mini module, connected via UART, retrieves pre-recorded audio files stored on a microSD card corresponding to classified gestures. A 3W mini speaker outputs the spoken response, providing immediate auditory feedback.

3.4 Power Management Layer

Each glove is powered by a 9V battery regulated using an AMS1117 voltage regulator. The ESP32-S3 and sensors operate at 3.3V, while the DFPlayer Mini and speaker receive a regulated 5V supply. This ensures stable and safe operation of all modules.

3.5 System Workflow

The operational flow is as follows:

1. The user performs an ISL gesture.
2. Flex sensors and the MPU6050 capture finger bending and motion dynamics.
3. The ESP32-S3 preprocesses and segments sensor data into temporal windows.
4. The embedded GRU model classifies the gesture.
5. The DFPlayer Mini retrieves the corresponding audio file.
6. The speaker outputs the translated speech.

4 Proposed Methodology

The proposed methodology integrates synchronized multi-sensor acquisition, temporal modeling, and embedded deep learning inference to enable real-time gesture recognition on resource-constrained hardware. The primary contribution lies in the structured multi-modal temporal representation and an embedded-aware learning pipeline optimized for sequential hand-gesture modeling.

4.1 Sensor Data Collection

Two instrumented gloves—a master (left hand) and a slave (right hand)—capture complete bimanual gesture dynamics. Each glove contains five flex sensors and an MPU6050 inertial measurement unit. The slave glove performs only data acquisition and transmits timestamped sensor readings to the master using ESP-NOW. All preprocessing, synchronization, temporal segmentation, and inference are executed on the master glove to maintain centralized feature construction and temporal consistency.

Data Acquisition and Transfer Workflow

- **Local Sampling:** Flex sensors are sampled via ESP32 ADC channels, while the MPU6050 provides tri-axial acceleration and angular velocity through the I²C interface. The sampling rate is selected to preserve gesture transitions as structured time-series signals suitable for recurrent modeling.
- **Wireless Transmission:** The slave glove transmits timestamped raw packets using ESP-NOW for low-latency peer-to-peer communication, enabling synchronized dual-hand acquisition.
- **Synchronization and Windowing:** The master glove aligns incoming streams using timestamps and segments the synchronized data into fixed-length temporal windows. Instead of handcrafted feature extraction, calibrated raw signals are directly arranged into ordered sequences for data-driven temporal modeling.

- **Signal Conditioning:** Moving-average filtering reduces high-frequency noise and ADC artifacts. Sensor-specific calibration and normalization minimize inter-user variability. At each timestep, a unified feature vector is formed by concatenating flex and IMU measurements from both gloves in a fixed order for GRU input.
- **Embedded-Aware Design:** Buffering, window management, and preprocessing are implemented using fixed-size memory allocation and minimal floating-point overhead to ensure deterministic real-time execution on the ESP32 platform.

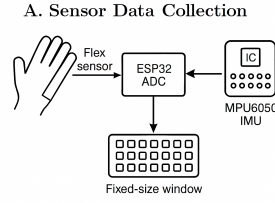


Fig. 2. Sensor Data Collection process showing flex sensors, ESP32 ADC, and MPU6050 IMU data flow into fixed-size windows.

4.2 Data Preprocessing

Sensor readings are normalized using calibration offsets to reduce inter-user variability. Low-pass filtering suppresses noise while preserving temporal transitions. The resulting temporal windows are formatted as structured sequences representing finger articulation and hand motion.

4.3 Model Inference

An LSTM model was initially evaluated for sequential gesture modeling; however, its higher parameter count and computational complexity led to increased memory usage and slower inference on the ESP32-S3. A 1D-CNN architecture provided faster parallel computation but showed weaker performance in modeling long-term temporal dependencies.

The GRU architecture offered a balanced trade-off between temporal modeling capability and computational efficiency. With fewer gating operations than LSTM, GRU reduces memory footprint and inference latency while preserving sequential learning capacity. Based on classification stability, memory usage, and real-time feasibility on embedded hardware, GRU was selected for deployment.

The trained GRU model was quantized and embedded on the ESP32-S3 using TensorFlow Lite Micro, enabling fully on-device real-time gesture classification.

4.4 Audio Mapping and Playback

Each predicted gesture label corresponds to a pre-recorded audio file stored on a microSD card within the DFPlayer Mini module. Upon classification, the ESP32 issues a playback command, and the connected speaker outputs the translated speech.

5 Model Training and Deployment

The gesture recognition model was developed using a dataset collected from six participants performing 10 commonly used Indian Sign Language (ISL) gestures. Each gesture was repeated 40 times per participant, resulting in 2880 samples. Each sample consisted of a 2-second sequence captured at 50 Hz with a 0.2-second delay, producing 10 time-steps per gesture. Each time-step contained 22 features: 10 flex sensor values and 12 IMU values (3-axis accelerometer and 3-axis gyroscope from both gloves).

5.1 Dataset Preparation

Signals were normalized using sensor-specific calibration offsets and filtered using low-pass filtering to reduce flex sensor noise and IMU drift. The dataset was shuffled and divided into:

- 70% training (2016 samples)
- 15% validation (432 samples)
- 15% testing (432 samples)

5.2 GRU Model Architecture

Three architectures—LSTM, 1D-CNN, and GRU—were evaluated considering classification stability, computational complexity, memory footprint, and embedded inference feasibility on the ESP32-S3.

LSTM demonstrated strong temporal modeling but incurred higher parameter count and computational overhead due to multiple gating operations, leading to increased memory usage and slower inference on constrained hardware. The 1D-CNN model provided faster parallel computation but showed comparatively limited ability to model long-term temporal dependencies in sequential sensor data.

GRU provided a balanced trade-off between temporal modeling capability and efficiency. With reduced gating complexity compared to LSTM, GRU achieved lower memory consumption and faster inference while maintaining consistent classification performance. Its recurrent structure also better preserved sequential dependencies compared to 1D-CNN, making it more suitable for time-series gesture modeling on embedded hardware.

The final deployed architecture consisted of:

- GRU layer with 48 hidden units
- Dropout layer (rate 0.15)
- Dense classification layer with 12 neurons (Softmax)

This configuration ensured stable generalization while remaining compatible with real-time inference constraints on microcontroller-class hardware.

5.3 Training Configuration

The model was trained using TensorFlow 2.12 with:

- Optimizer: Adam
- Learning rate: 0.0008
- Loss: Sparse categorical cross-entropy
- Epochs: 60
- Batch size: 32

Training converged without overfitting due to dropout regularization.

5.4 Model Deployment

The trained GRU model was quantized to int8 using post-training quantization, reducing model size from 148 KB to 41 KB. The model fits within the 512 KB RAM of the ESP32-S3 and achieves an average inference time of 92–118 ms per window at 160 MHz. Deployment was performed using TensorFlow Lite Micro with a 60 KB tensor arena.

6 Hardware Components

The system consists of two smart gloves: a **Master Glove** (left hand), responsible for on-device inference and audio playback, and a **Slave Glove** (right hand), responsible for sensing and wireless transmission.

Master Glove (Left Hand)

The master glove integrates an **ESP32-S3** microcontroller for embedded GRU inference using TensorFlow Lite Micro. It is connected to five **flex sensors** for finger-bend detection and an **MPU6050** IMU for wrist orientation and motion tracking. Gesture predictions are converted to speech using a **DFPlayer Mini** MP3 module interfaced via UART, with output delivered through a compact **3W speaker**.

Slave Glove (Right Hand)

The slave glove uses a standard **ESP32 (ESP32-DevKit V1)** for data acquisition and wireless transmission. It contains the same configuration of flex sensors and MPU6050 for bimanual gesture capture. Sensor readings are transmitted to the master glove using the ESP-NOW protocol for low-latency peer-to-peer communication.

Power Management

Each glove is powered by a 9V battery regulated using an **AMS1117** voltage regulator. The ESP32 boards and sensors operate at **3.3V**, while the DFPlayer Mini and speaker require a stable **5V** supply. This regulated power distribution ensures reliable operation while maintaining lower power consumption on the slave glove, which does not perform inference or audio processing.

7 Results and Evaluation

This section evaluates the developed smart glove system in terms of prototype implementation, gesture recognition accuracy, latency, wireless transmission, power consumption, and user feedback.

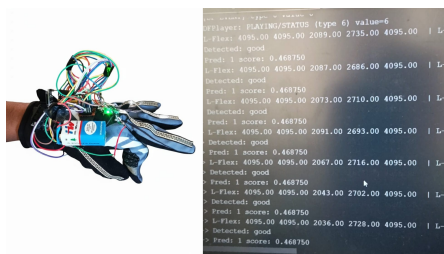


Fig. 3. Glove prototype being used for real-time performance testing, showing live sensor readings and gesture prediction output.

7.1 Prototype Implementation

Fig. 4 shows the completed master and slave glove prototypes used during testing. Each glove integrates five flex sensors, an MPU6050 IMU, an ESP32 microcontroller, power supply circuitry, and signal wiring, reflecting the actual hardware configuration used for data collection and real-time inference.

7.2 Gesture Recognition Performance

The final GRU model was evaluated on a test set of 12 ISL gestures with the following results:

- Training accuracy: 95.3%
- Validation accuracy: 92.1%
- Test accuracy: 90.8%

Misclassifications primarily occurred between gestures with similar finger configurations or overlapping wrist orientations, such as “Hello” and “Thank You,” and dynamic gestures such as “Yes” and “OK.”



Fig. 4. Master (left) and slave (right) glove prototypes with flex sensors, ESP32 modules, IMU, and wiring.

7.3 Real-Time Latency Evaluation

End-to-end latency was measured from gesture completion to audio playback. Average component delays were:

- Preprocessing: 18–25 ms
- GRU inference (ESP32-S3): 92–118 ms
- DFPlayer Mini command delay: 38–45 ms

The overall response time was approximately **155–185 ms**, enabling natural conversational interaction.

7.4 Wireless Transmission Performance

Sensor data was transmitted from the slave glove to the master using ESP-NOW. Measured delays were:

- Line-of-sight: 3.8 ms average
- Indoor usage: 6.2 ms average

Packet loss remained below 1.7% during continuous streaming and did not significantly impact recognition performance.

7.5 Battery and Power Consumption

Using 9V rechargeable batteries, runtime was:

- Master glove: 3.7–4.2 hours
- Slave glove: 5.4–6.1 hours

Higher consumption on the master glove is attributed to on-device inference and audio playback.

7.6 User Study

A user study involving 12 participants, including two experienced ISL users, evaluated practical usability. Average ratings (out of 5) were:

- Accuracy satisfaction: 4.3
- Ease of use: 4.5
- Comfort and wearability: 3.9
- Responsiveness: 4.4

Participants reported better reliability for dynamic gestures compared to subtle static finger poses and suggested improved wiring integration for enhanced comfort.

8 Conclusion and Future Work

This paper presented SignSpeak, a wearable system for real-time gesture-to-speech translation for individuals with hearing and speech impairments. By integrating flex sensors, an MPU6050 IMU, and an embedded GRU-based TFLite Micro model on the ESP32-S3, the system performs on-device gesture recognition without external computation.

The prototype achieved 90.8% test accuracy with an end-to-end latency below 185 ms, demonstrating practical feasibility for real-time communication using a dual-glove embedded architecture.

Future Work

Future improvements include expanding the ISL gesture vocabulary, enabling continuous sentence-level translation, integrating optional Bluetooth/Wi-Fi connectivity, incorporating vibration or visual feedback for bidirectional interaction, redesigning hardware using compact PCBs for improved ergonomics, exploring flexible sensing materials, and supporting multilingual audio output.

8.1 Limitations of the Presented Work

Despite promising performance, several limitations remain:

- The current system supports only 12 predefined ISL gestures, limiting vocabulary scalability and continuous translation capability.
- The dataset was collected from six participants, which may not fully capture inter-user variability in hand size, gesture style, and motion dynamics.
- Long-term usage may introduce sensor drift, calibration inconsistencies, and wearability concerns due to discrete wiring and modular hardware design.
- Gesture outputs rely on pre-recorded audio rather than dynamic text-to-speech synthesis, limiting phrase flexibility.
- Continuous operation is constrained by battery life (approximately 4 hours for the master glove).

Addressing these limitations will support broader deployment and improved robustness for real-world applications.

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